tscircuit-boy-advance

Hardware to prepare - quantities for ONE device



7 x M2 x 4

LCD (5) + speaker (2)

Countersunk / Phillips PH0



2 x M2 x 5

Battery-holder keepers

Countersunk / Phillips PH0



4 x M3 x 6

PCB + 4 nylon washers

Pan head / Phillips PH1



4 x M2.5 x 16

Close the two shells

Countersunk / Phillips PH1

Heat-set inserts: match BOTH outside diameter and length



9 x M2

OD 3.6 x length 3.0

LCD + speaker + holder



4 x M3

OD 4.6 x length 5.7

PCB mounts



4 x M2.5

OD 4.6 x length 4.0

Shell closure

PCB washers: nylon, 3.2 ID x 7 OD x 0.5 thick.

For five devices: multiply every quantity above by five. Screw sketches are illustrative, not to scale.

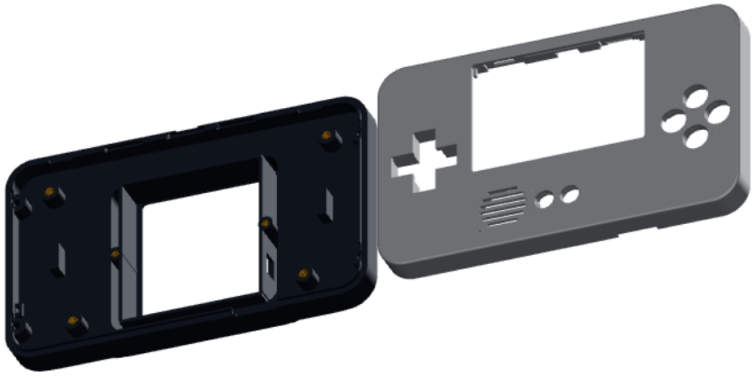
Countersunk length includes the head. Pan-head length excludes the head. Install inserts in empty shells; test the clearance coupon first.

Do not substitute M3 OD5 x L4 inserts for OD4.6 x L5.7 without checking the printed mounts.

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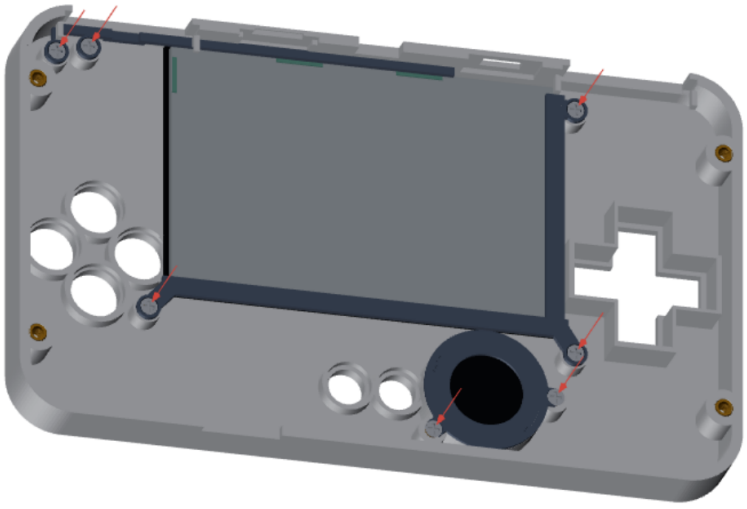
Build the front shell

1. Heat-set the inserts



Install all inserts while the shells are EMPTY. Front: 7 M2 + 4 M2.5. Rear: 2 M2 + 4 M3. Keep them square and flush; let them cool.

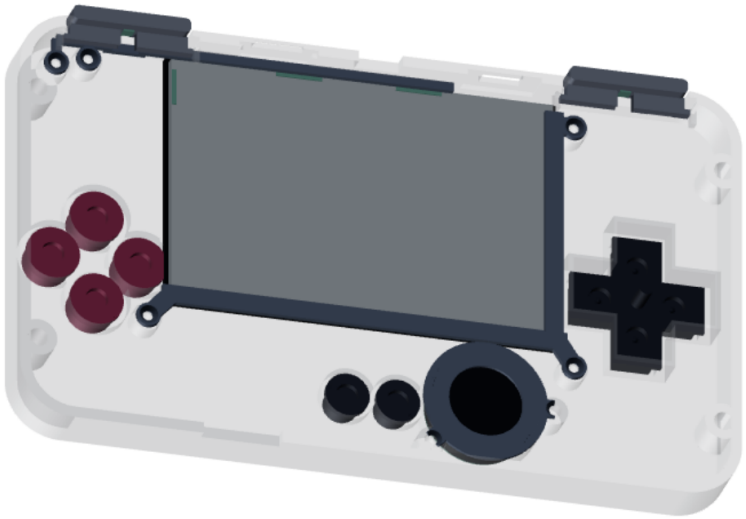
2. Fit display and speaker



Seat the display on soft inactive-border foam. Fit LCD retainers with 5 M2 x 4 screws; speaker ring with 2 M2 x 4. Tighten gently. Keep buttons out until done.

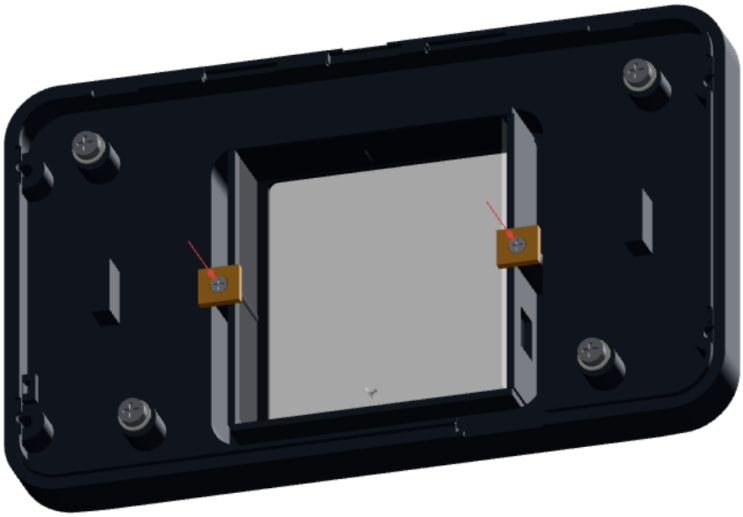
Display warning: the no-touch panel may be thinner than the modelled glass stack. Check fit and foam before tightening.

3. Load the buttons



Insert D-pad, A/B/X/Y and Start/Select from inside the front shell. Fit shoulders through their lead-ins: 0.7 mm outward, slide up 6 mm, then seat inward.

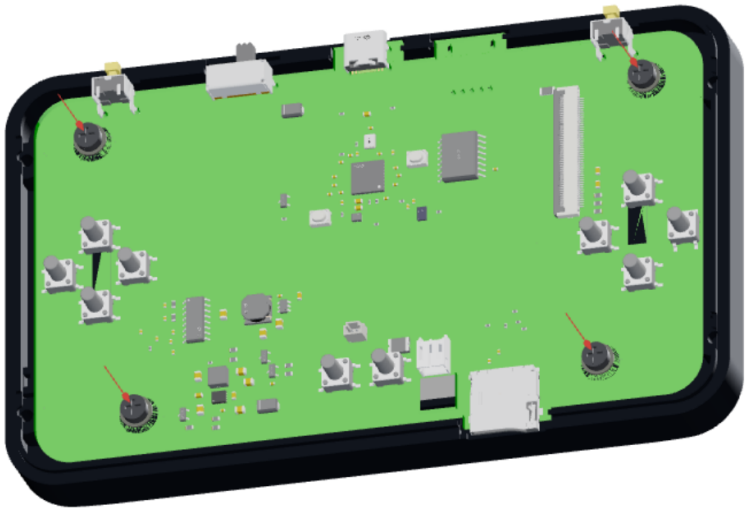
4. Secure the battery holder



Leave AA cells out. Open cell side faces the battery door. Fit both printed keepers with 2 M2 x 5 screws. The holder mounts to the REAR SHELL, never the PCB.

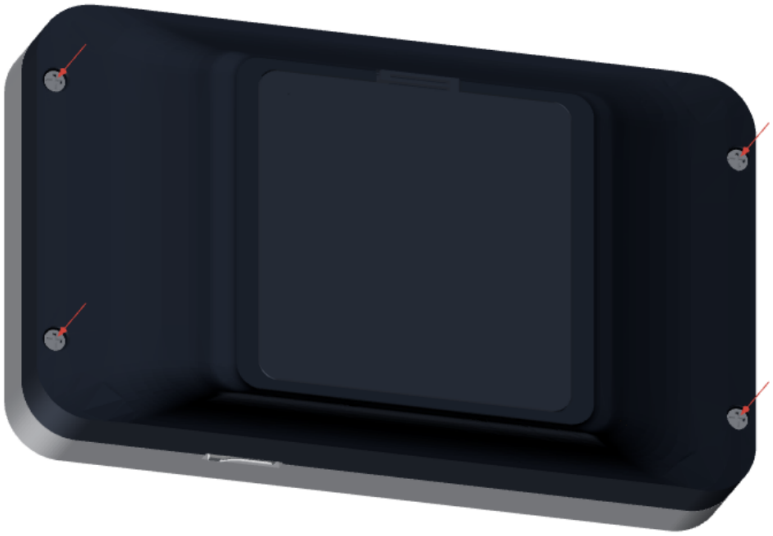
Ensure every cap returns freely. Do not force the shell shut or glue the moving guides.

5. Mount PCB and connect cables



Fit 4 M3 x 6 pan-head screws, each with a nylon washer. Battery: PHR-2; speaker: SHR-02V-S-B. Insulate splices. Connect and lock display flex before closing.

6. Close, check, then add batteries



Keep wires and flex clear of caps and screw tips. Close with 4 M2.5 x 16 screws. Check every control. Verify polarity; install 4 AA cells last. Door uses a plastic latch, no screws.

J_BAT: pin 1 positive, pin 2 ground. J_SPK: pin 1 L+, pin 2 L-; neither is ground. USB does not charge AA cells.